1.6 SAY'S LAW OF MARKET

The Say's law of market has been a basic pillar of classical economic theory. According to this law, every supply generates its own demand. An implication of the above is that if a commodity is produced, it will create sufficient income for the owners of factors such as land, labor and capital to purchase the produced commodity. After all, the price of the commodity is divided into its components such as rent, wages and profits. Hence, this income will be sufficient to realize the price of the commodity.

According to the Say's law, enough income will always exist to purchase the entire produced output, so that there will not be any surplus left under the laissez-faire orientation. Using this logic, it can be concluded that there should not be any major depression. If it occurs, it must be caused by some interference with the free flow of commodities and money. The Say's Law thus supported the laissez-faire orientation of classical economics. The law is heavily dependent on the assumption of perfect flexibility in prices. Under this assumption, whatever quantity is produced can be sold in the market. In the short run, supply (or demand) may exceed demand (or supply). Such discrepancy is adjusted automatically, and instantaneously, through a decline (or rise) in prices. Such instantaneous adjustment exists for all commodities and factor inputs. Says' law is also regarded as a natural consequence of perfect competition.

The proponents of the Say's law believe that the law is valid both in a barter economy as well as a money economy. The law states that income received from the production of output is always spent on aggregate demand, i.e., consumption and investment. In other words, the law suggests that money is never hoarded and that the money or expenditure stream (MV) remains neutral. There is no doubt that the law is valid under a barter economy where production was primarily for consumption, i.e., whatever is produced is exchanged for goods and services. But one may doubt its validity in the present era, when production is based on future expectations and anticipations of demand; there is bound to be some overproduction. It is often said that the Say's law is valid if a substantial portion of production is used for consumption and the rest, if saved, is invested.

Check Your Progress 2

1) Describe how output and employment are determined in the classical model.

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2) What are the implications of the quantity theory of money?

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3) What is the relevance of the Say's law?

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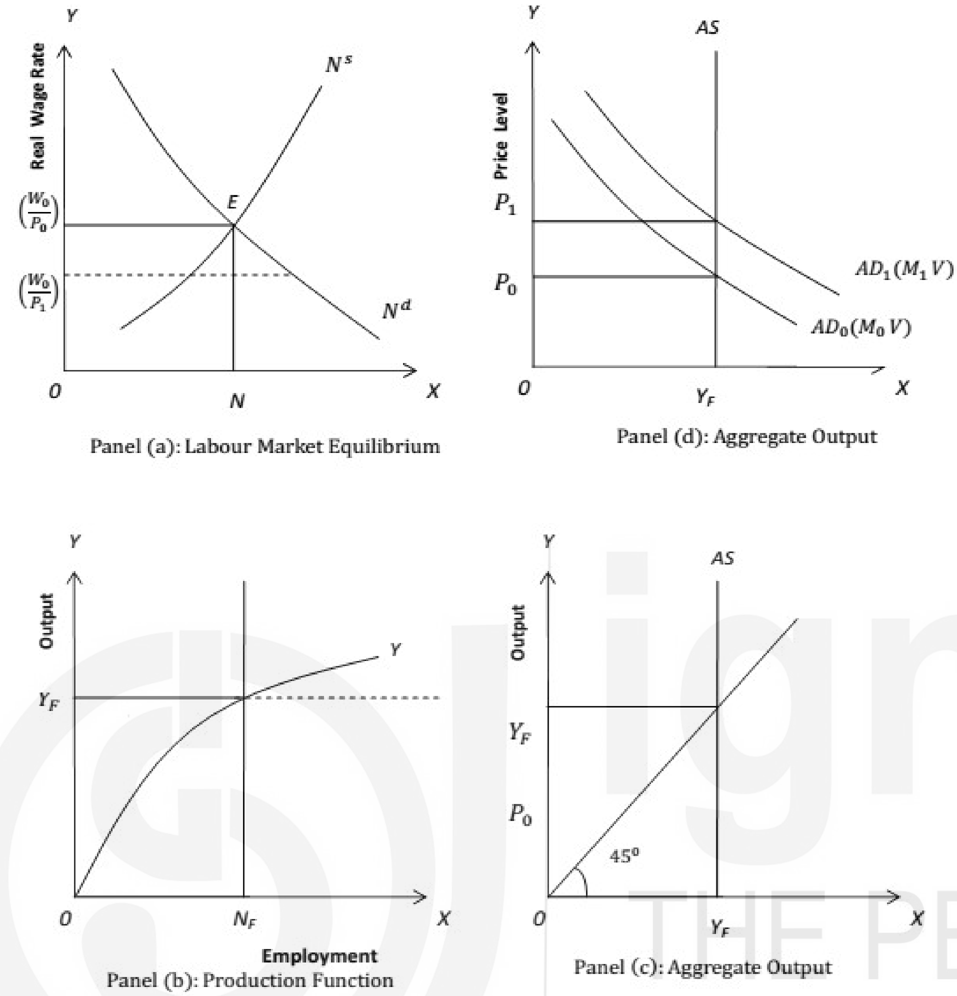
1.4. CLASSICAL DICHOTOMY

From the quantity theory of money (QTM) we find the relationship between money supply and price level. We discuss below the effect of changes in money supply on other macroeconomic variables.

1.4.1 AD-AS Equilibrium

In an economy the real variables are real GDP, real capital stock and employment. These variables measure certain physical quantity. For instance, real GDP is defined as the quantity of goods and services produced in a specific period, such as a year, quarter or month. Similarly, real capital stock is defined as the quantity of machines and structures available at a specific time. The nominal variables are expressed in terms of monetary value, for example, the price level and the wage of a person in rupees.

Classical theory of output and employment suggests that the changes in the quantity of money affects nominal variables such as prices, nominal wages and nominal income. Changes in quantity of money, however, does not affect real variables. Classical economists provide an argument that real supply-side factors such as population, technology, capital stock, state of technology, marginal physical product of labour, and households' preferences regarding work and leisure determine real output and employment. The assumption of flexibility in prices leads to the self-adjustment of markets. Flexibility of prices and wages suggests that the changes in money supply affects the price level and nominal values such as money wages and nominal interest rates while real variables remain unaffected. Such independence of real economic variables from changes in money supply and nominal variables is known as 'classical dichotomy'.



Panel (a): Labour Market Equilibrium
Panel (d): Aggregate Output
Panel (b): Production Function
Panel (c): Aggregate Output

Fig. 1.5: Classical Model of Output Determination

1.4.2 Neutrality of Money

An implication of classical dichotomy is that 'money is neutral'. We have illustrated 'neutrality of money' graphically in Fig. 1.5. There are four panels in Fig. 1.5. Suppose money supply in an economy is equal to M_0 at a specific time. The corresponding aggregate demand curve is AD_0 as seen in panel (d) of Fig. 1.5. Its interaction with the aggregate supply curve AS decides the price level P_0. Corresponding to the price level P_0, the labour market determines the equilibrium money wage rate W_0. Hence, the real wage rate is equal to W_0/P_0 and equilibrium level of employment is N_F as seen in Panel (a) of Fig. 1.5. Given the production function (see Panel (b)), with the equilibrium level of employment N_F the aggregate output is Y_F. Let us assume that the monetary authority of the country expands its money supply from M_0 to M_1 which results in an upward shift in the aggregate demand curve to AD_1 as seen in Panel (d) of Fig. 1.5. As a result, the price level increases from P_0 to P_1. This causes the real wage to fall from (W_0/P_0) to (W_0/P_1) as seen in Panel (a) of Fig. 1.5. At the real wage decreases to (W_0/P_1), the demand for labour increases. Thus, labour demand exceeds labour supply. According to classical theory, this causes the nominal wage rate to increase to W_1 (in equal proportion to the increase in price level) such that the original level of real wage is maintained (that is, $W_1/P_1 = W_0/P_0$). Thus, there is no change in the equilibrium level of employment N_F. (See panel (a) of Fig. 1.5). At the earlier wage rate and employment level, the real output is not affected, as seen in panel (b) of Fig. 1.5. To conclude, when there is an expansion of money supply, the nominal wage rate and price level increase without affecting the levels of real wage, employment and output. Such independence of real economic variables from changes in money supply is called neutrality of money.

There is a crucial limitation to the neutrality of money. It is a basic result derived from the full-employment equilibrium that is based on the full flexibility of prices. If increase in money supply (resulting in higher prices) had no real effects, then inflation would not have been a serious concern in an economy. Inflation, however, is a serious problem as has many adverse effects such as decrease in quality of life of people, decrease in economic growth, etc. Thus, measures are taken to control inflation and maintain price stability in an economy.

1.4.3 Saving-Investment Equilibrium

You should note that classical theory emphasized on the function of money that it is a medium of exchange. According to the classical school money is demanded for transaction purposes only. Hence, supply of and demand for money in the classical theory do not determine the equilibrium rate of interest. An increase in the quantity of money does not affect the real rate of interest. Thus, there is no change in the level of saving and investment (see, Fig. 1.6). This shows that when money supply rise, it does not disturb the capital market equilibrium (or saving-investment equality) and the level of full-employment equilibrium remains unchanged.

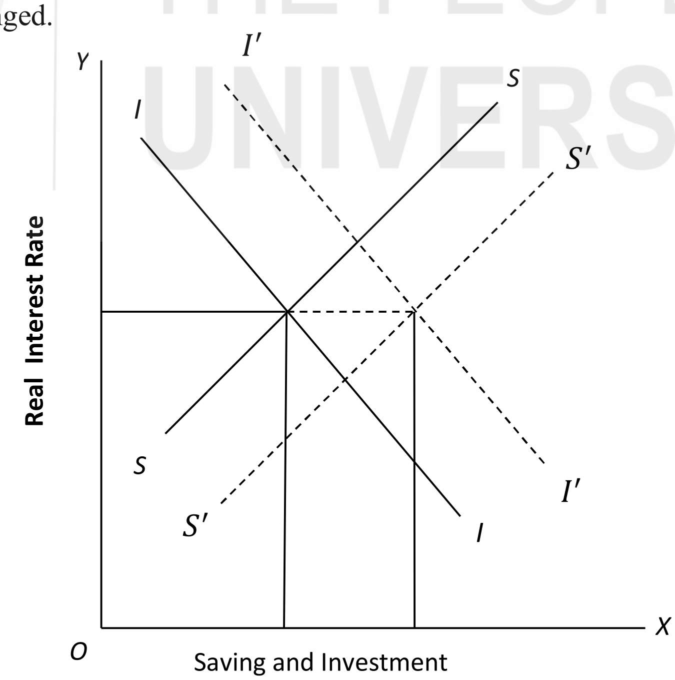


Fig. 1.6: Capital Market Equilibrium

An increase in money supply will lead to an increase in prices. Due to the higher prices, nominal investment expenditure will increase. However, according to classical theory, such increase will be proportional to the increase in prices. Therefore, investment expenditure in real terms will not change. This is explained diagrammatically in Fig. 1.6. We measure real interest rate on the y-axis and nominal value of saving-invest on the x-axis. The increase in money supply causes the supply curve of nominal saving to shift downward to the right from SS to $S'S'$. Simultaneously, the investment demand curve will shift upward from II to $I'I'$ by the same amount. It implies that the interaction of $S'S'$ and $I'I'$ determines the interest rate without affecting real saving and real investment at the higher price level too.

1.5 LONG RUN VS. SHORT RUN

In general, supply and demand fluctuate for various reasons, which affects the level of output. There is considerable difference between short-run and long-run fluctuations in output. Most economists believe that the behavior of processes is different between the short run and the long run. In the long run, prices are flexible, which may be affected by changes in supply or demand. In the short run, most of the prices are "sticky'' at the pre-determined level. As the behaviour of prices in the short run is different from that in the long run, the effects of economic events and policies vary over different time horizons. Classical theory is considered to be applicable in the long run. In the short run, we may need a different analytical framework. In Unit 2 we discuss the Keynesian model, which is more applicable in the short run.

Let us explain the price behavior in both the short-run and long-run in Fig. 1.7. Aggregate supply is the total amount of goods and services that firms sell in an economy. Aggregate demand is the total quantity of goods and services that are purchased. In a standard AS-AD model, output (Y) is presented on the X-axis and price (P) is on the Y-axis. The long run aggregate supply curve (AS_{LR}), which denotes the potential output (full employment output) of the economy is given by a vertical line at Y^*. The short-run supply curve is upward sloping (AS_0). The interaction of supply and demand determines the equilibrium level of output (Y_0). In the short run, an outward shift in the supply curve (from AS_0 to AS_1) results in rising output (from Y_0 to Y_1) and falling prices (from P_0 to P_1). An outward shift in the AD curve (from AD_0 to AD_1) also results in rising output (from Y_0 to Y_2) but rising prices (from P_0 to P_2).

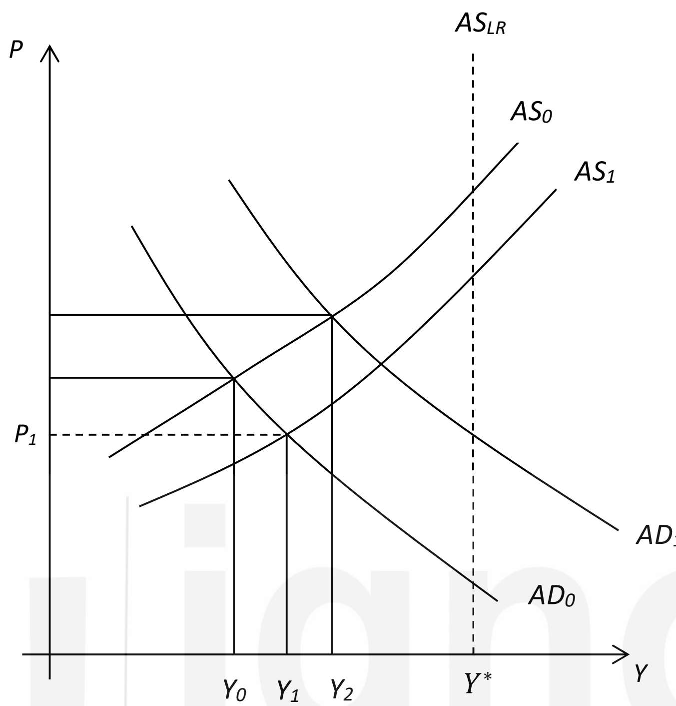


Fig. 1.7: Determination of Prices in the Long-Run and Short-Run

Let us look into certain other situations. Short-run fluctuations in nominal variables usually result in a change in the output level, as macroeconomic variables not fully flexible. An increase in money supply, for example, will shift the AD curve to the right (i.e., aggregate demand will increase). This will increase prices and income. In contrast, a decrease in money supply will result in a shift in the AD curve to the left, which will decrease both prices and income. According to the classical theory an increase in money supply leads to increased prices without affecting the real output. This may be applicable in the long run, not in the short run.

Check Your Progress 3

1) What is meant by classical dichotomy?

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2) Why is money considered to be neutral? What are its implications?

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3) Is classical theory applicable in the short run?

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1.6 LET US SUM UP

In this unit, we discussed different schools of macroeconomic thought that have evolved over time to respond to certain events in the world economy. Relying on the assumption of full flexibility of wages and prices, classical theory concludes that the economy always tends to equilibrium at the full employment level of output. Also, the capital market always ensures that there is enough demand in the market. The classical quantity theory of money determines the level of prices and wage rates without affecting the real economic variables. Classical theory advocates that policy intervention cannot influence output levels in an economy as it is self-adjusting. In the classical framework money considered to be neutral in the sense that it does not affect real variables.

1.7 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

1) In Section 1.3 we have given the important features of classical theory. Go through it and answer.

2) New-classical economics considers a general equilibrium framework, microfoundations and rational expectations.

Check Your Progress 2

1) Go through Sub-Section 1.4.1. Refer to Fig. 1.5 and describe the process.

2) The quantity theory of money (MV = PY) establishes the relationship between money supply, velocity of money, price level and output. When the velocity of money and the output level remain constant, an increase in money supply will lead to increase in prices.

3) According to the Say's law, supply creates its own demand. It rules out the possibility of unemployment in an economy.

Check Your Progress 3

1) It refers to the classical proposition nominal variables do not influence real variables in an economy. It arises from the classical assumptions of perfect competition and full flexibility in prices and wages.

2) Classical economists emphasized that money is demanded as a medium of exchange. An increase in money supply leads to proportionate increase in prices and wages; real variables such as output and employment remain unchanged.

3) Prices and wages may not be fully flexible. Rigidities in prices and wages may not lead to instantaneous adjustment in output and employment. Thus, classical theory may not be applicable in the short run.